CDN/WAF - edge

Region (3 AZs)

L7 Load Balancer

Kubernetes - autoscaled pods (HPA)

API Gateway x N

Reservation/Inventory pods x N

Product, Cart, Sale, Checkout, Payment, Order, Shipment, Notification pods

Prometheus, Grafana, Loki, Jaeger

Redis Cluster 3 shards + replicas

Postgres primary AZ-1

Kafka 3 brokers RF=3

Postgres sync replica AZ-2 + read replicas